

Scalability & Future Plan

Date	5 July 2026
Team ID	
Project Name	Voice-Based Concept Understanding Analyzer
Maximum Marks	1 Mark

Scalability & Future Plan

This document explains how the Voice-Based Concept Understanding Analyzer can be expanded to support a larger number of users, improved performance, cloud deployment, enhanced security, and additional Artificial Intelligence features in future releases.

Current System Limitations

S.No	Limitation	Impact	Priority
1	Supports only single-user execution.	Cannot handle multiple concurrent users.	High
2	Limited AI evaluation features.	Cannot evaluate advanced communication skills.	Medium
3	Uses local JSON storage	Limited scalability and centralized storage.	High

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user Streamlit application.	Deploy on cloud with multi-user support and load balancing.
2	Data Storage	JSON files.	Use PostgreSQL or Firebase for centralized storage.
3	Performance	CPU-based processing.	Optimize AI inference and enable GPU acceleration.
4	Security	Basic local authentication.	Add user authentication, encrypted storage and role-based access.

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud Deployment using Streamlit Community Cloud	1 Month	Online accessibility for users.
Phase 2	Database Integration (PostgreSQL/Firebase)	2 Months	Reliable and scalable data management.

Phase 3	Advanced AI-based Speech & Emotion Analysis	4 Months	Improved evaluation accuracy and detailed feedback.
Phase 4	Mobile Application & Multilanguage Support	6 Months	Increased accessibility and wider user adoption.